

E9: 309 ADL 16-12-2020

<http://leap.ee.iisc.ac.in/sriram/teaching/ADL2020/>

Recap from previous lectures

* Analyzing trained neural networks

- ✓ Hierarchical representations ✓
 - ◉ Maximizing activations ✓
 - ◉ Visualizing representations ✓
 - ◉ Reconstruction of input patterns from hidden layers. ✓

* Transferability of representations

Today's lecture

✦ Why models predict what they predict



Architecture updates for interpretability

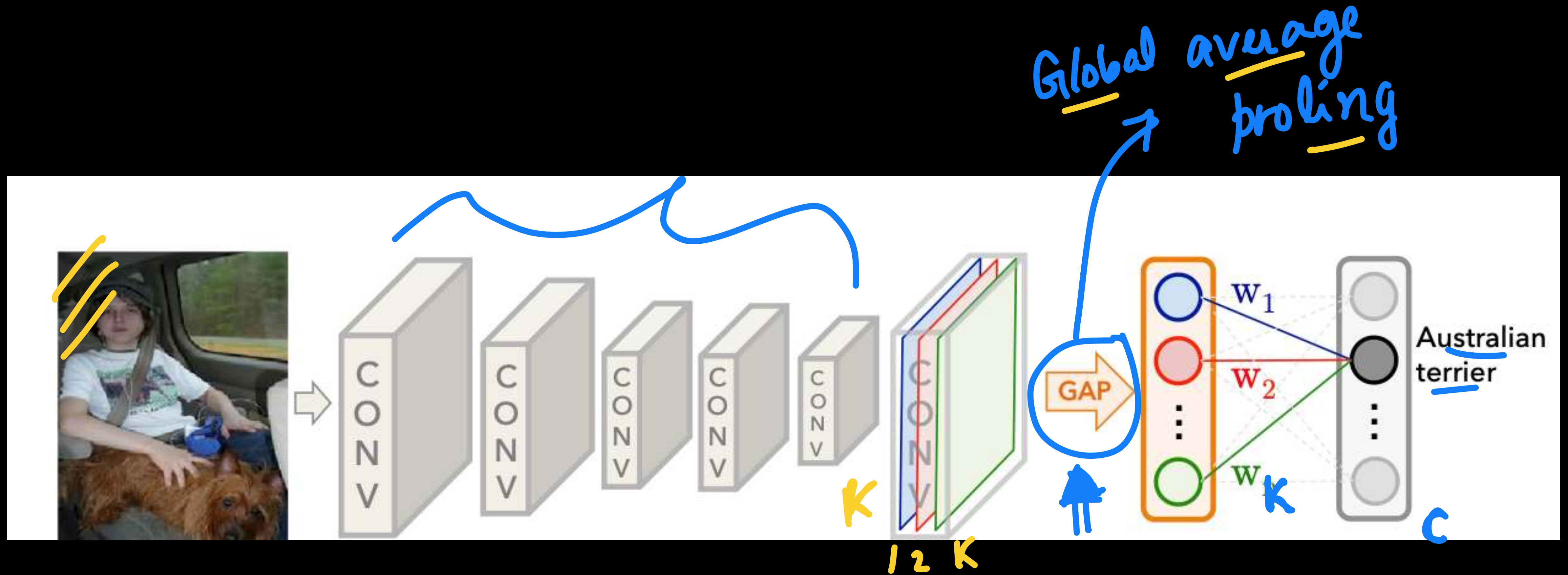
Learning Deep Features for Discriminative Localization

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2015



Learning the input pattern of a trained network



Architecture updates for interpretability

* Global average pooling

$$F^{k,L} = \sum_{i,j} \underline{f^{k,L}(i,j)}$$

k - index of feature map

$k = 1 \dots K$

L - last layer

* Last layer mapping to classes

$$\underline{a^{c,L}} = \sum_{k=1}^K \underline{w_c^{k,L}} \underline{F^{k,L}}$$

$$\hat{y} = \text{softmax}(\underline{a^L})$$

$$\underline{a^L} = \begin{bmatrix} a^{1,L} \\ a^{2,L} \\ \vdots \\ a^{C,L} \end{bmatrix}$$



Architecture updates for interpretability

*Rearranging the terms

$$1 \quad a^{c,L} = \sum_{k=1}^K w_c^{k,L} \sum_{i,j} f^{k,L}(i,j)$$

GAP

$$= \sum_{i,j} \sum_{k=1}^K w_c^{k,L} f^{k,L}(i,j)$$

Image

$a^{c,L}$ → Output activation

*Defining the map

$$m^c(i,j) = \sum_{k=1}^K w_c^{k,L} f^{k,L}(i,j)$$

class activation maps



Architecture updates for interpretability

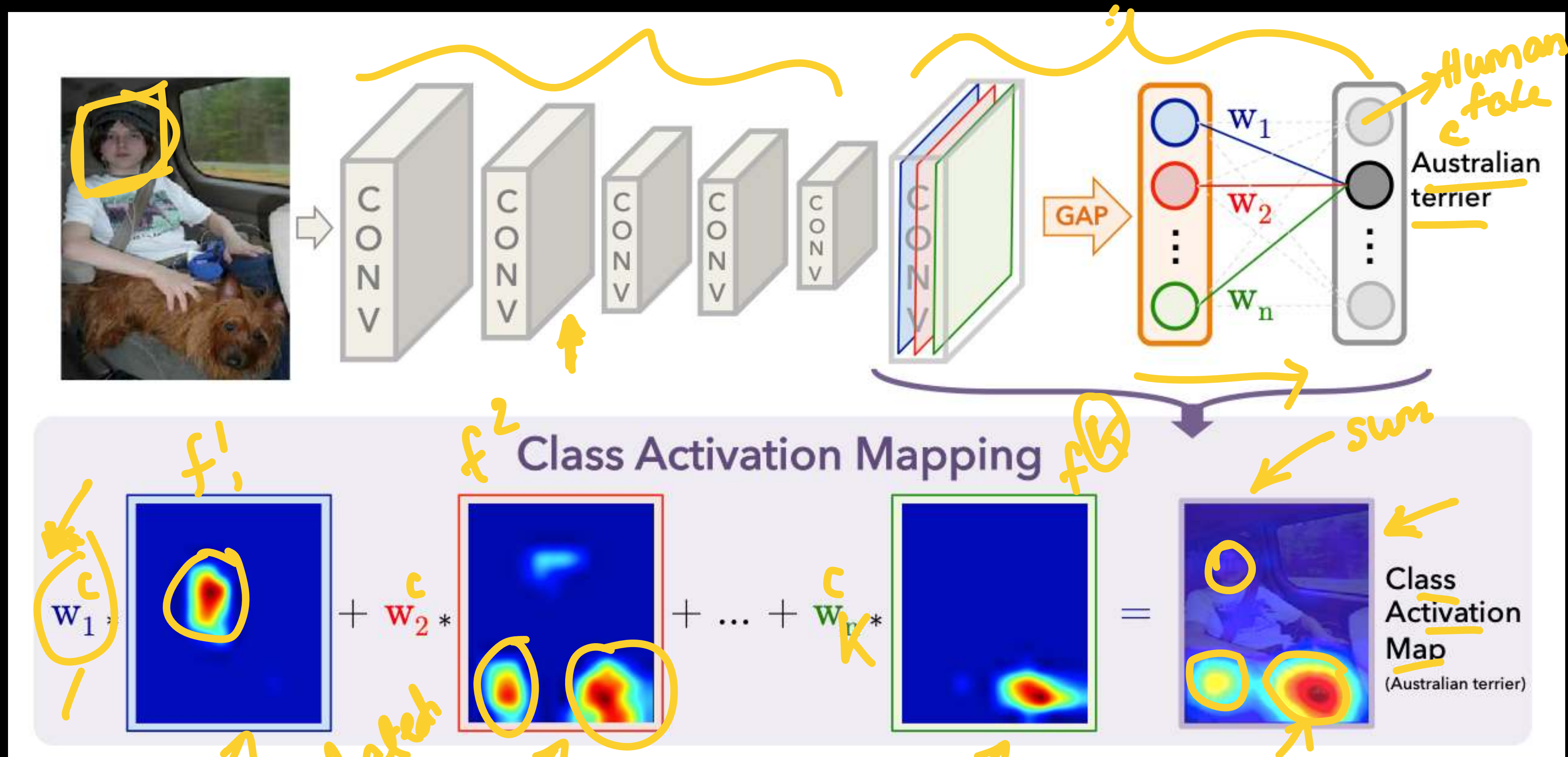
CAM

- * The maps denote spatial patterns that define how important a particular pixel is for that particular class.

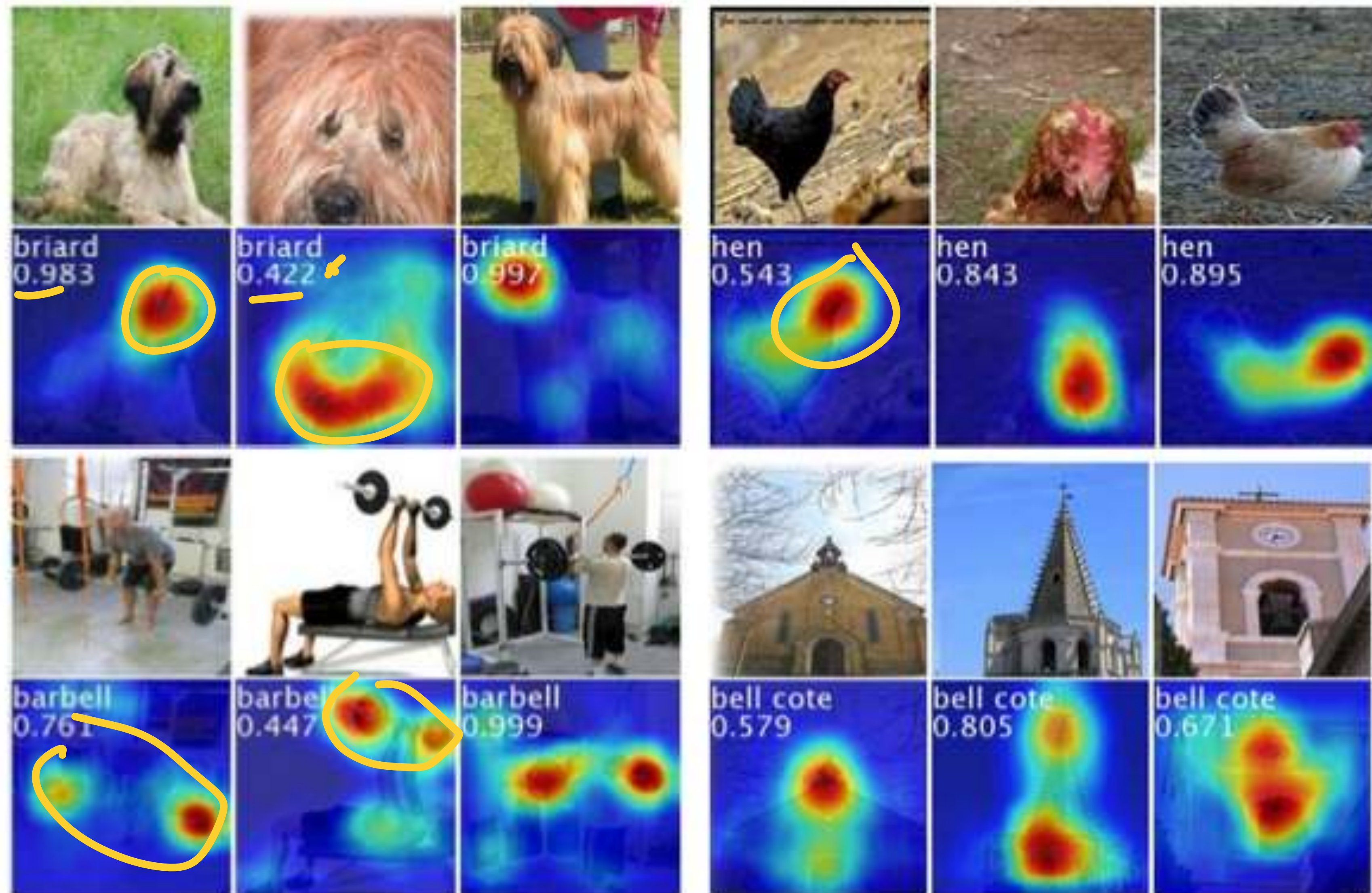


- * The map can be interpolated to the original input dimensions to visualize the pattern

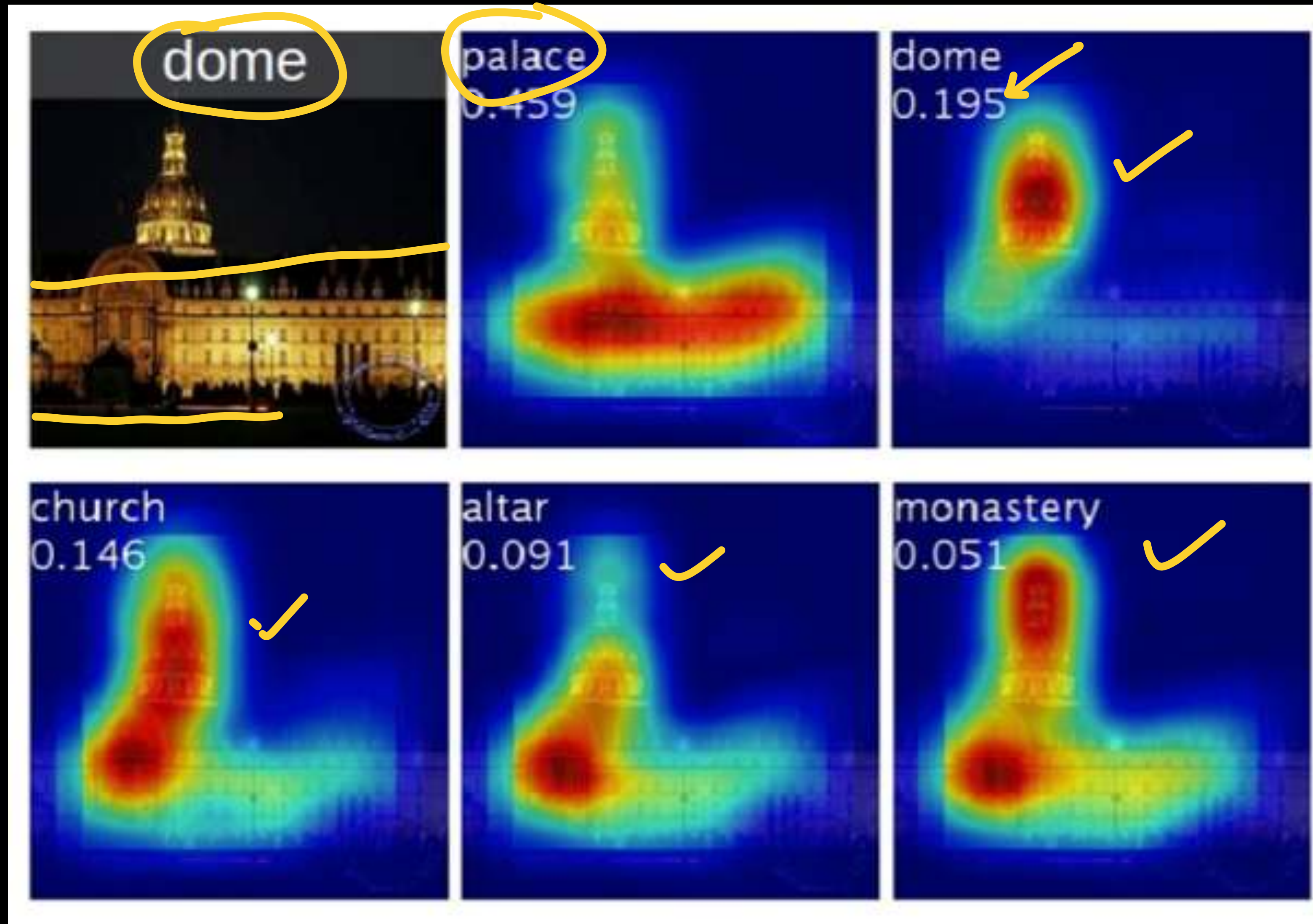
Architecture updates for interpretability



Architecture updates for interpretability



Architecture updates for interpretability



Architecture updates for interpretability

* Flaws in the approach

→ Requires Global average pooling based

→ The model may be inferior to the other models with fully connected layers after the CNN layers

Table 1. Classification error on the ILSVRC validation set.

Networks	top-1 val. error	top-5 val. error
VGGnet-GAP	33.4	12.2
GoogLeNet-GAP	35.0	13.2
AlexNet*-GAP	44.9	20.9
AlexNet-GAP	51.1 +	26.3 +
GoogLeNet	31.9	11.3
VGGnet	31.2	11.4
AlexNet	42.6	19.5



Grad-CAM : Visual Explanations

Selvaraju et .al (2016)
CVPR

Improving CAM without compromising architecture

- * Find the gradient of the output activation with respect to the feature maps of the last convolutional layer

$$\alpha_k^c = \frac{1}{S} \sum_{i,j} \frac{\partial a^{c,L}}{\partial f^{k,l}(i,j)}$$

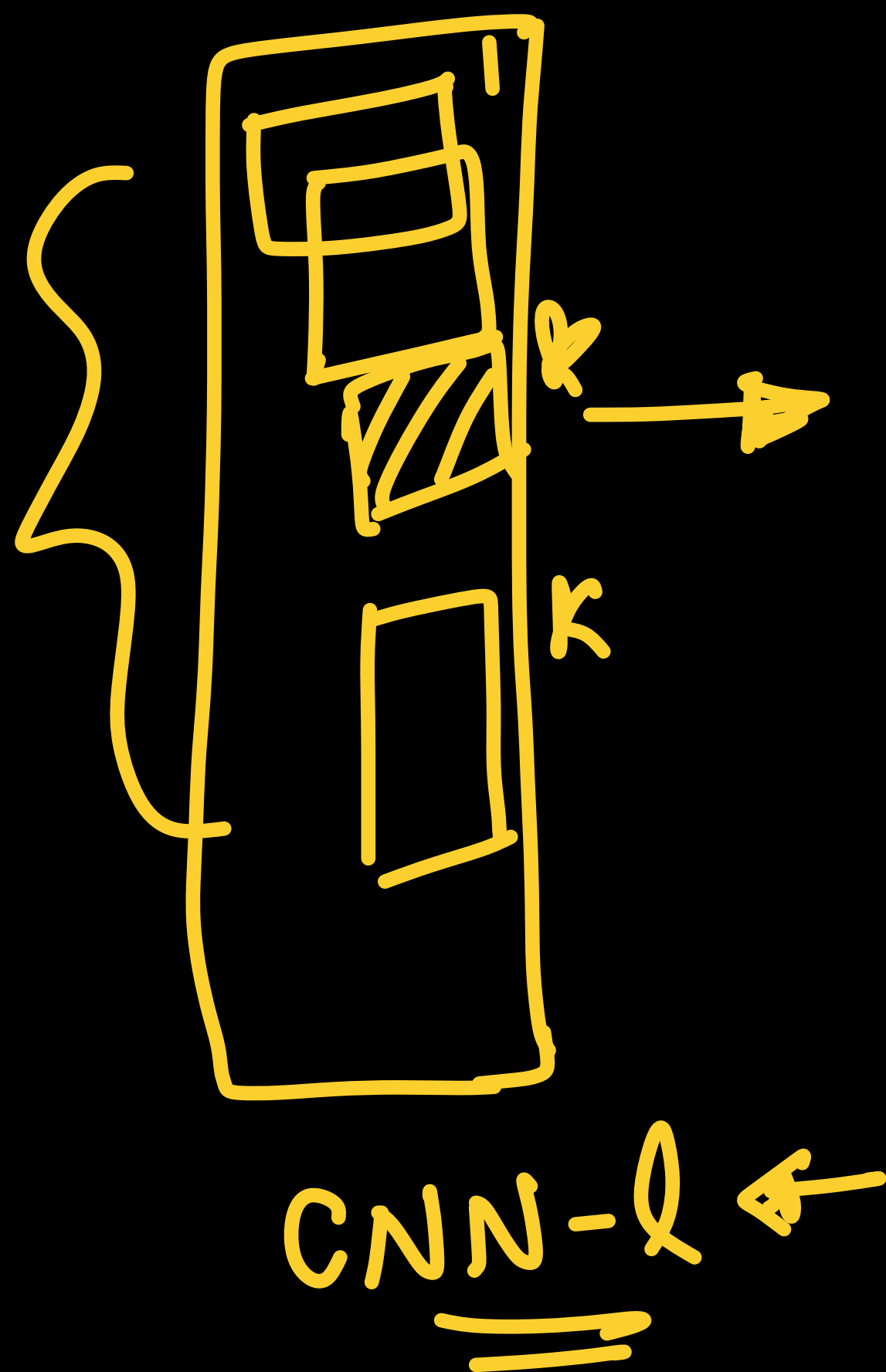
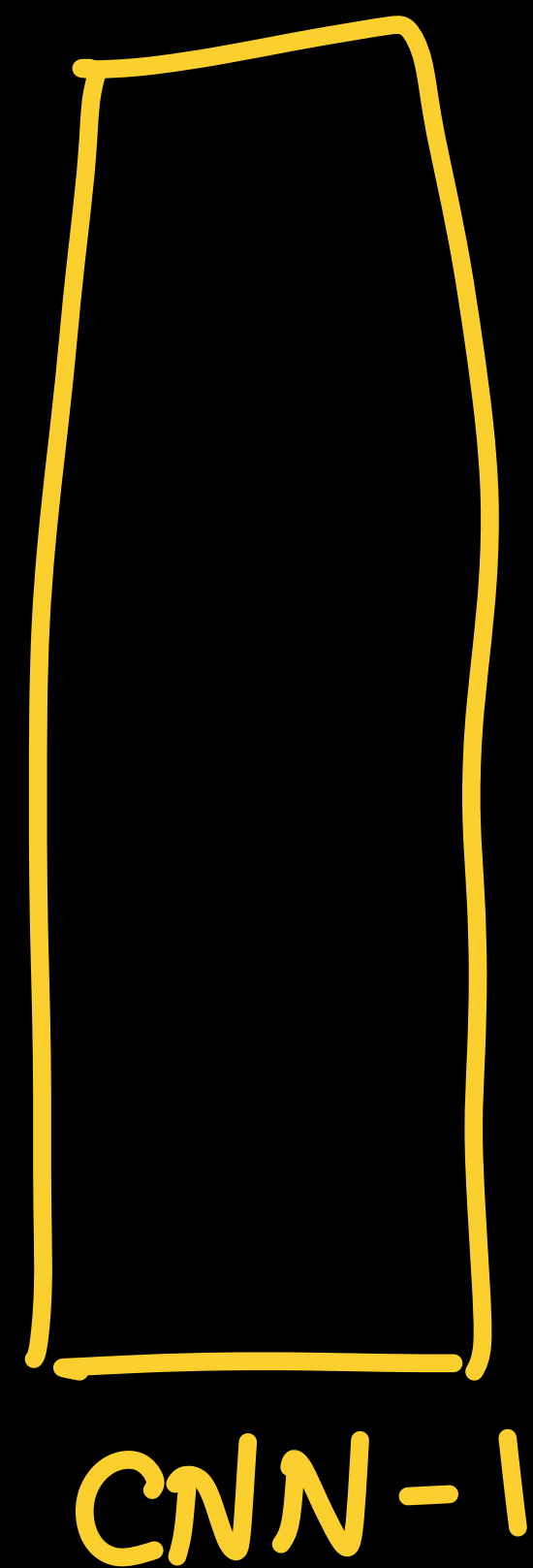
$$f^{k,l}(i,j)$$

- * Assumption - The feature maps from the last convolutional layer capture the spatial information as well as the semantic information required for classification.

$$S = \sum_{i,j} 1$$

l - last CNN layer
 k - feature map index





$$\frac{\partial a_c^L}{\partial f_k^L(i,j)}$$



Improving CAM without compromising architecture

* Gradient based activation maps (Grad-CAM)

$$\mathbf{m}^c = \text{ReLU} \left(\sum_k \alpha_k^c \mathbf{f}^{k,l}(i, j) \right)$$

Diagram illustrating the Grad-CAM formula with handwritten annotations:

- $\mathbf{m}^c$ is circled in yellow, with an arrow pointing to it from the left.
- ReLU is underlined in yellow, with an arrow pointing to it from below.
- The summation $\sum_k$ is underlined in yellow.
- α_k^c is underlined in yellow, with an arrow pointing to it from above.
- $\mathbf{f}^{k,l}(i, j)$ is underlined in yellow, with an arrow pointing to it from above.
- A bracket under the entire term $\alpha_k^c \mathbf{f}^{k,l}(i, j)$ is labeled "Interpolate" in yellow.
- An arrow points from the text "CAM output" in yellow to the entire expression.

* Multiply the gradient based contribution to the individual pixels of the feature maps

* The ReLU operation preserves only the pixels that have a positive influence on the output activations.

→ The negative pixels may belong some other class category



Relation between CAM and Grad-CAM

- * If the last convolutional layer is followed by a global average pooling (GAP) and softmax layer then, Grad-CAM gives

$$F^{k,L} = \sum_{i,j} \mathbf{f}^{k,L}(i,j) \quad \text{GAP}$$

- * The derivative output activation w.r.t GAP output

$$\frac{\partial a^{c,L}}{\partial F^{k,L}} = w_c^{k,L}$$

$$\frac{\partial a^{c,L}}{\partial \mathbf{f}^{k,L}(i,j)} = w_c^{k,L}$$



Relation between CAM and Grad-CAM

- * If the last convolutional layer is followed by a global average pooling (GAP) and softmax layer then, Grad-CAM gives

$$\underline{\underline{\alpha_k^c}} = \frac{1}{S} \sum_{i,j} \frac{\partial a^{c,L}}{\partial \mathbf{f}^{k,l}(i,j)} \quad \longrightarrow \quad \alpha_c^k = w_c^{k,L}$$

- * Grad-CAM generalizes the CAM framework to neural networks that can have convolutional networks followed by other architectures.



Visualizing Grad-CAM outputs



Tiger-cat



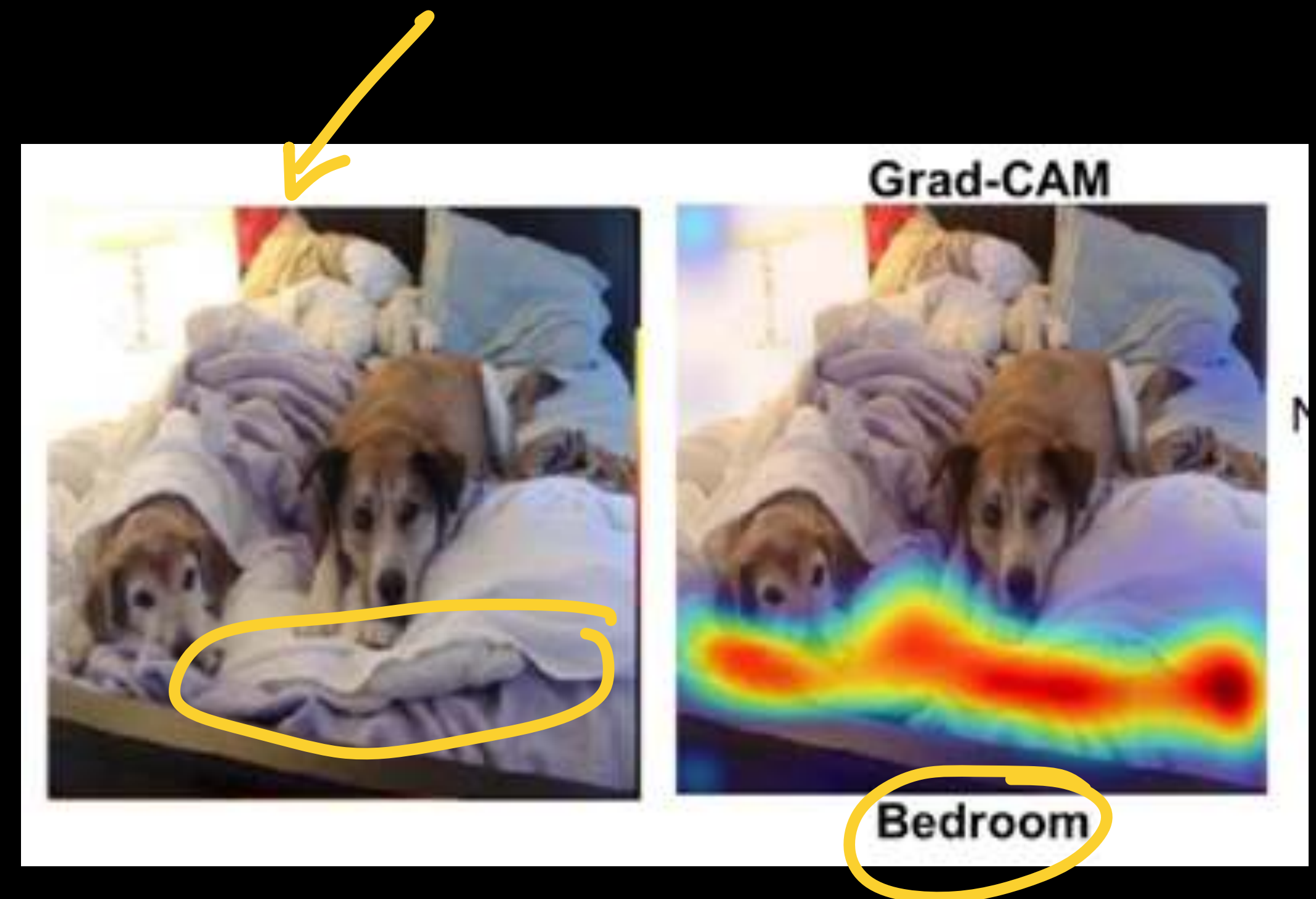
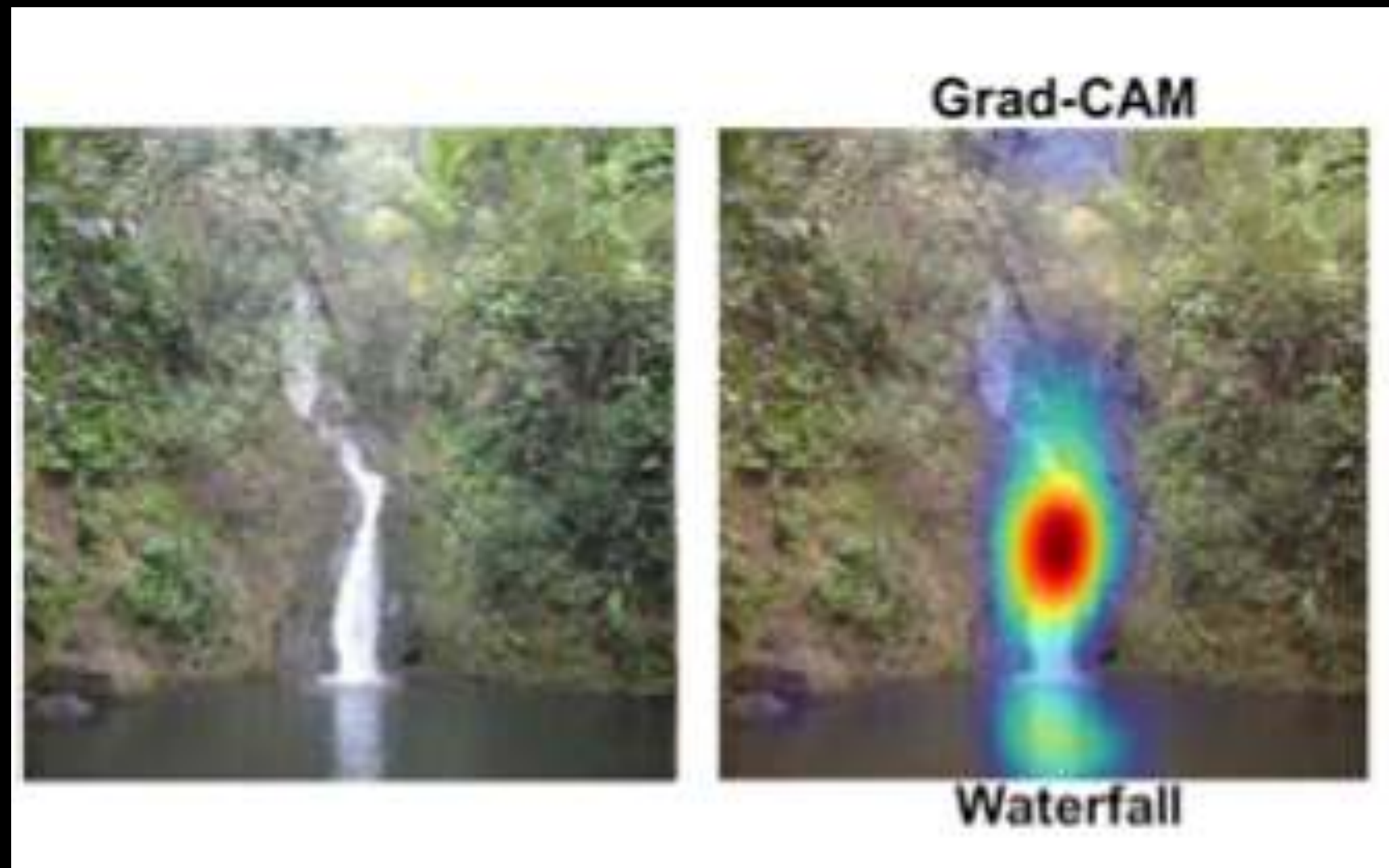
(c) Grad-CAM 'Cat'



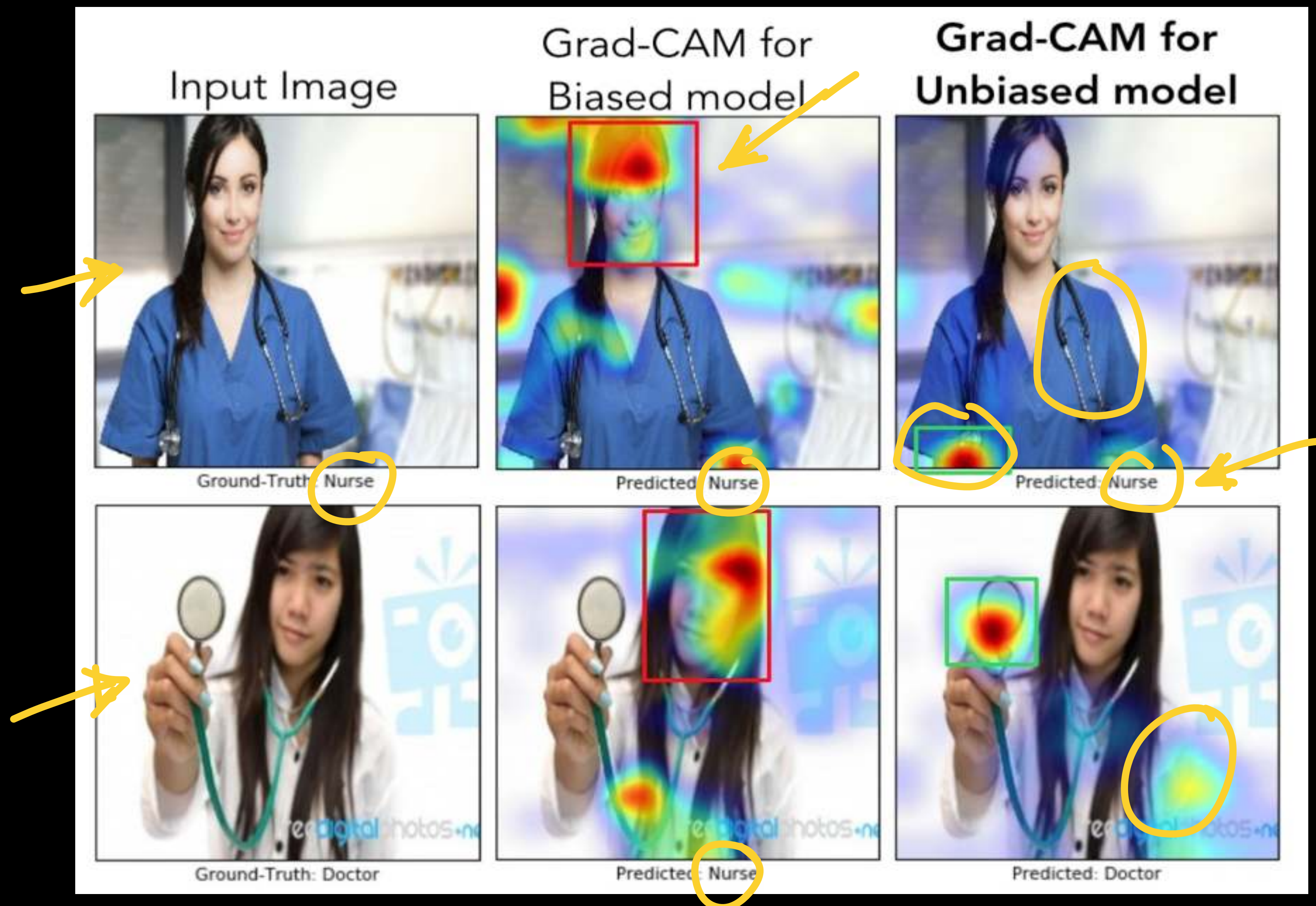
(i) Grad-CAM 'Dog'

↑
100 ←
101 ←
→ (90) ↑

Grad-CAM in image captioning

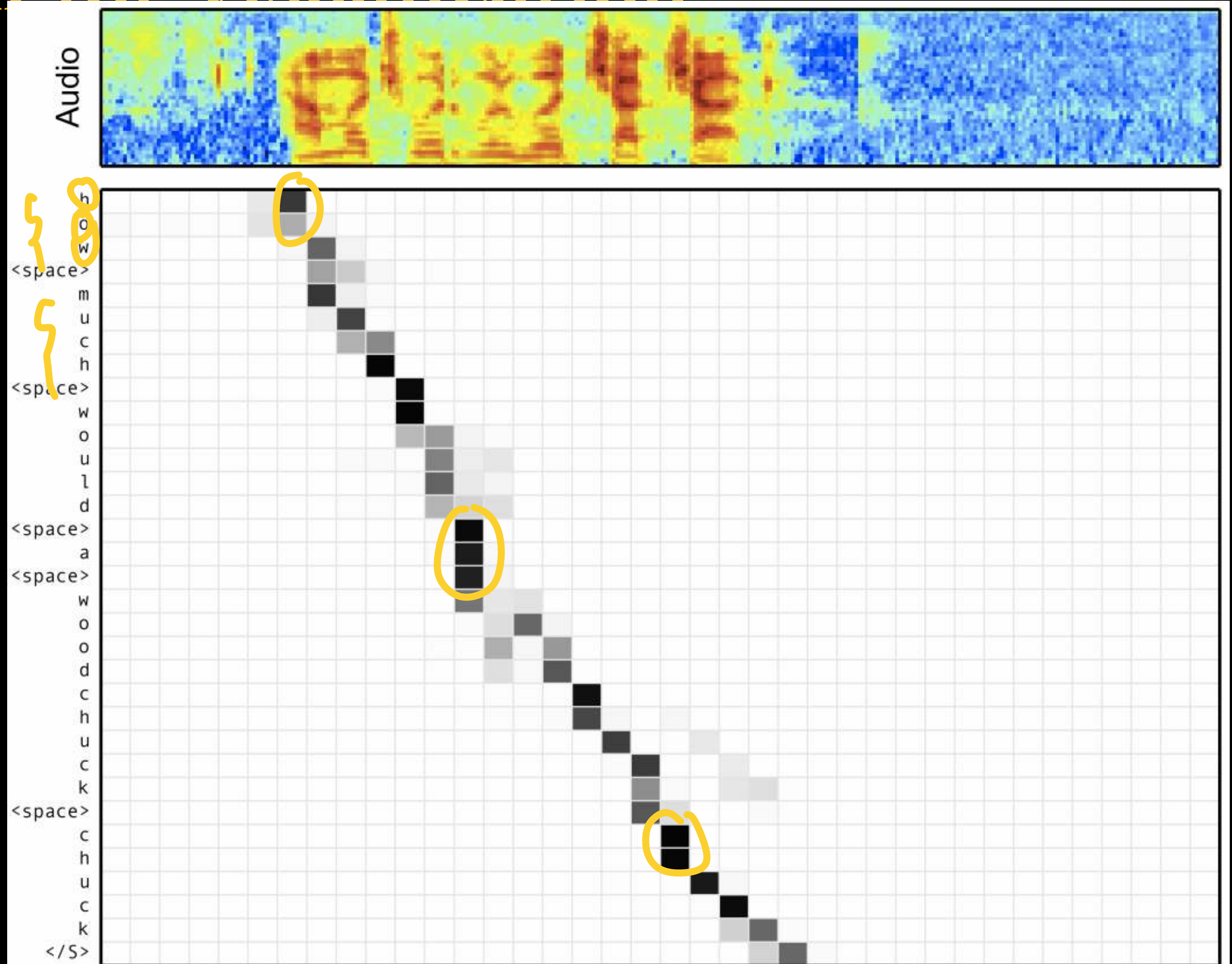


Grad-CAM for identifying model biases



Using attention for visualization

Audio



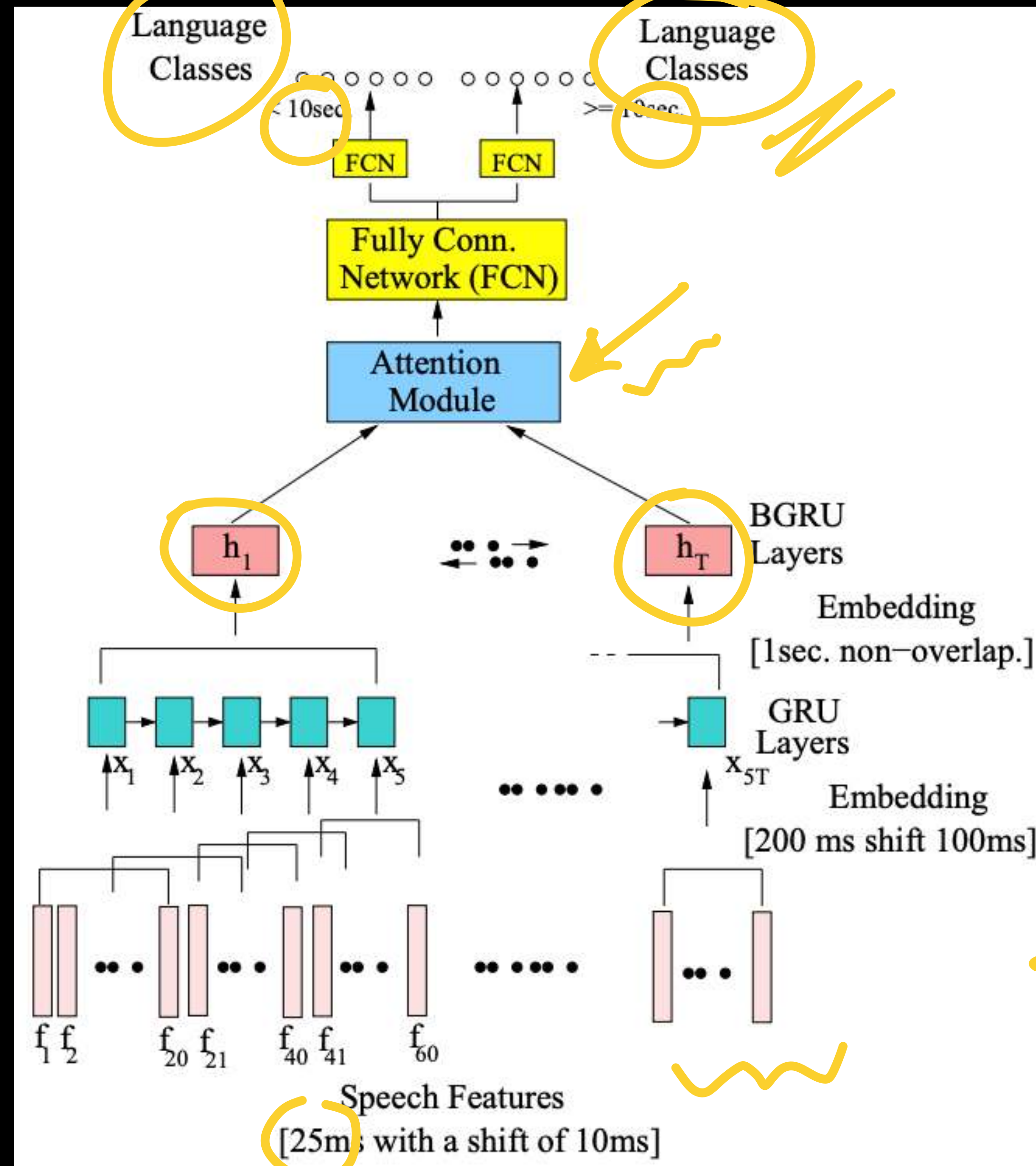
Using attention mechanism for explainability

Towards Relevance and Sequence Modeling in Language Recognition

Bharat Padi, Anand Mohan and Sriram Ganapathy, *Senior Member, IEEE*



Using attention mechanism for explainability



Speech

Eng

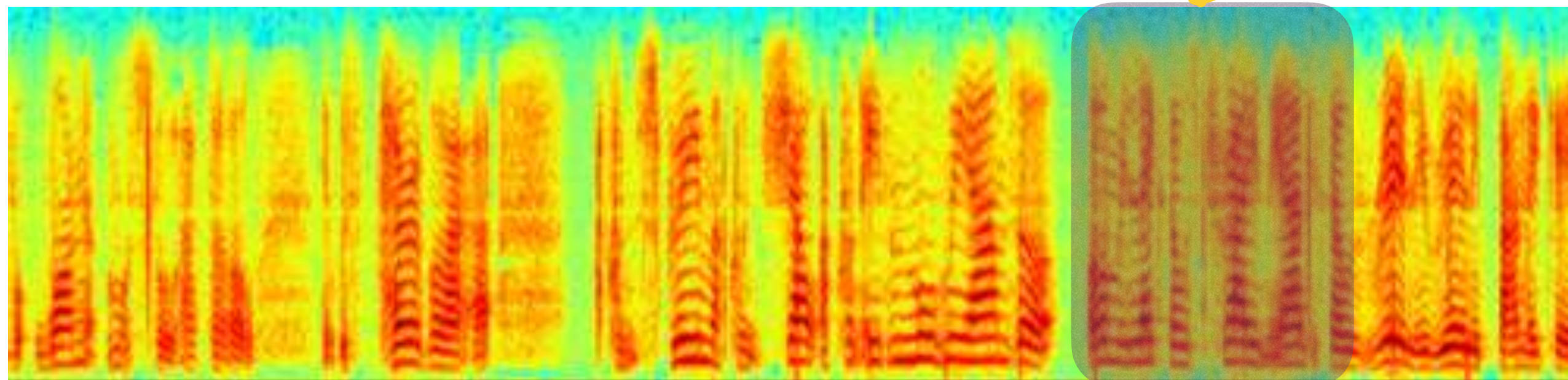
1/5

1/5

1/40

Using attention mechanism for explainability

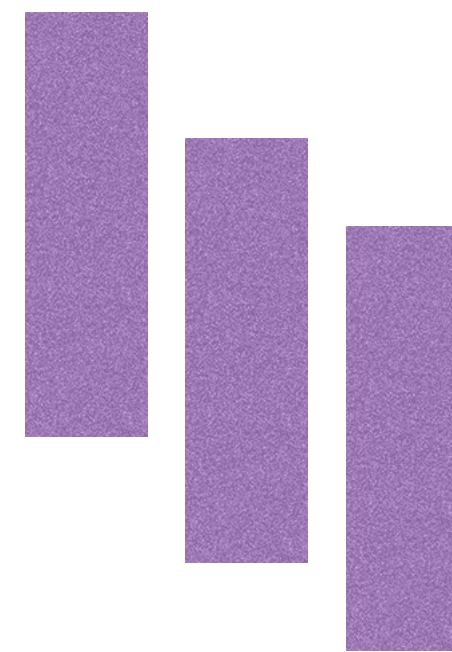
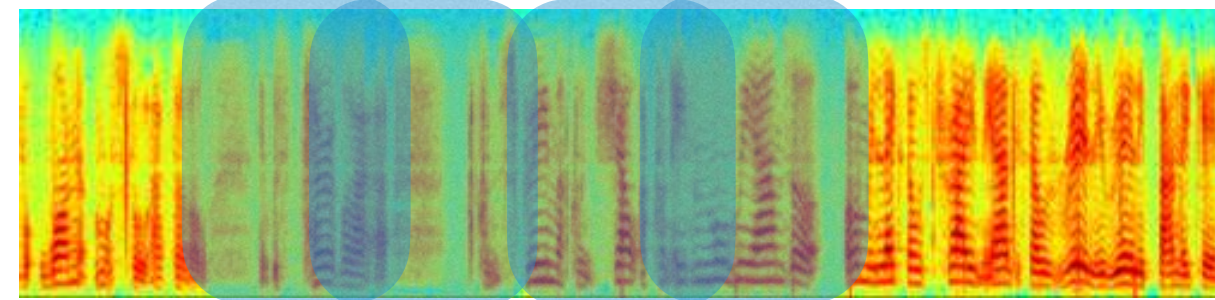
- Certain regions of the audio signal may have more information for the task than the rest.
- ✓ May also have more signal quality than rest.
- ✓ For example, language identification involving Eng-UK v/s Eng. US.



- Current models - use the information from audio with uniformity.

Using attention mechanism for explainability

- Derive short segment i-vectors



Sequence to label
model
using **attention**



- Attention weighs the importance of each short-term segment feature for the task.

Attention Weight

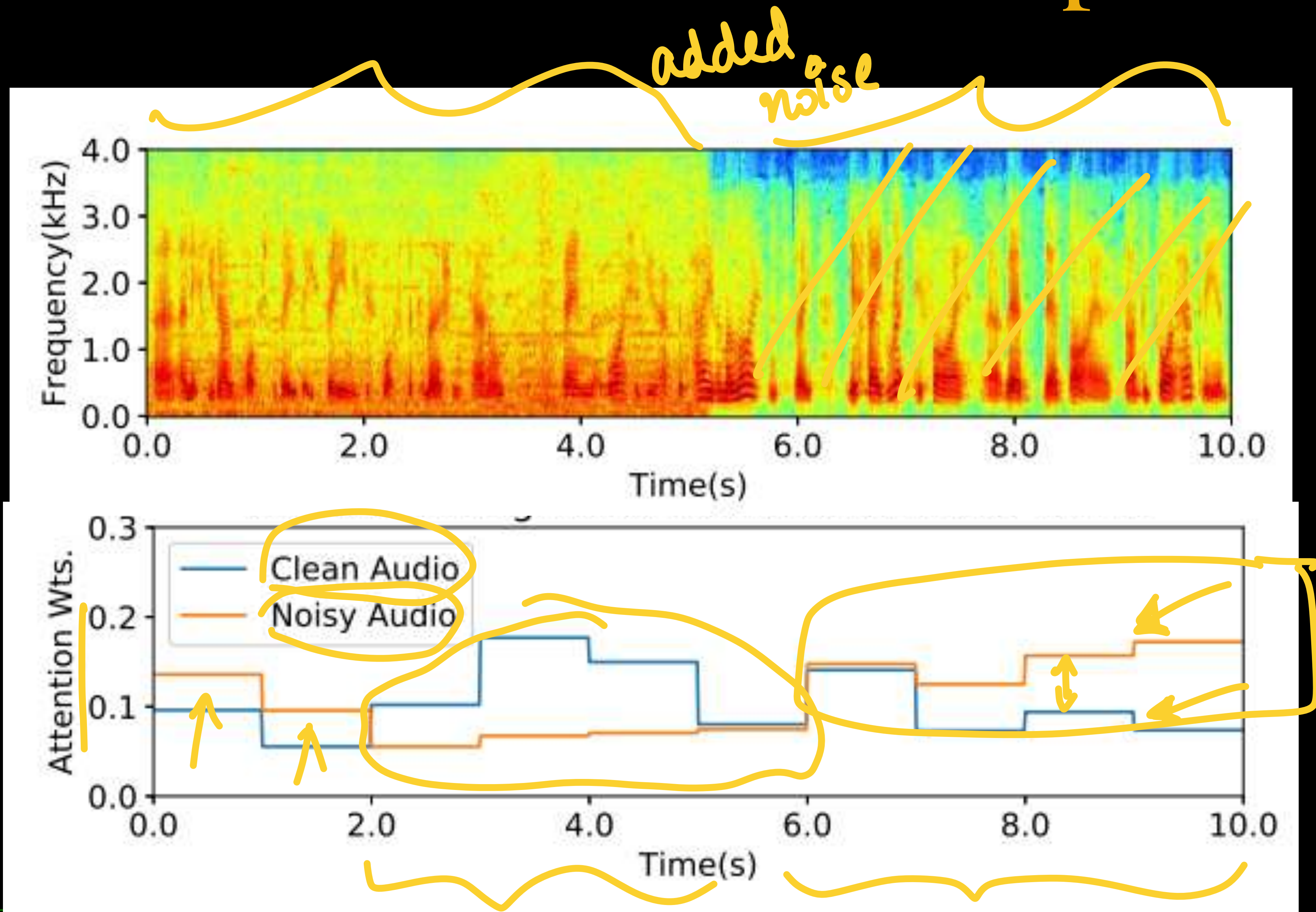
0-3s

3s-4s

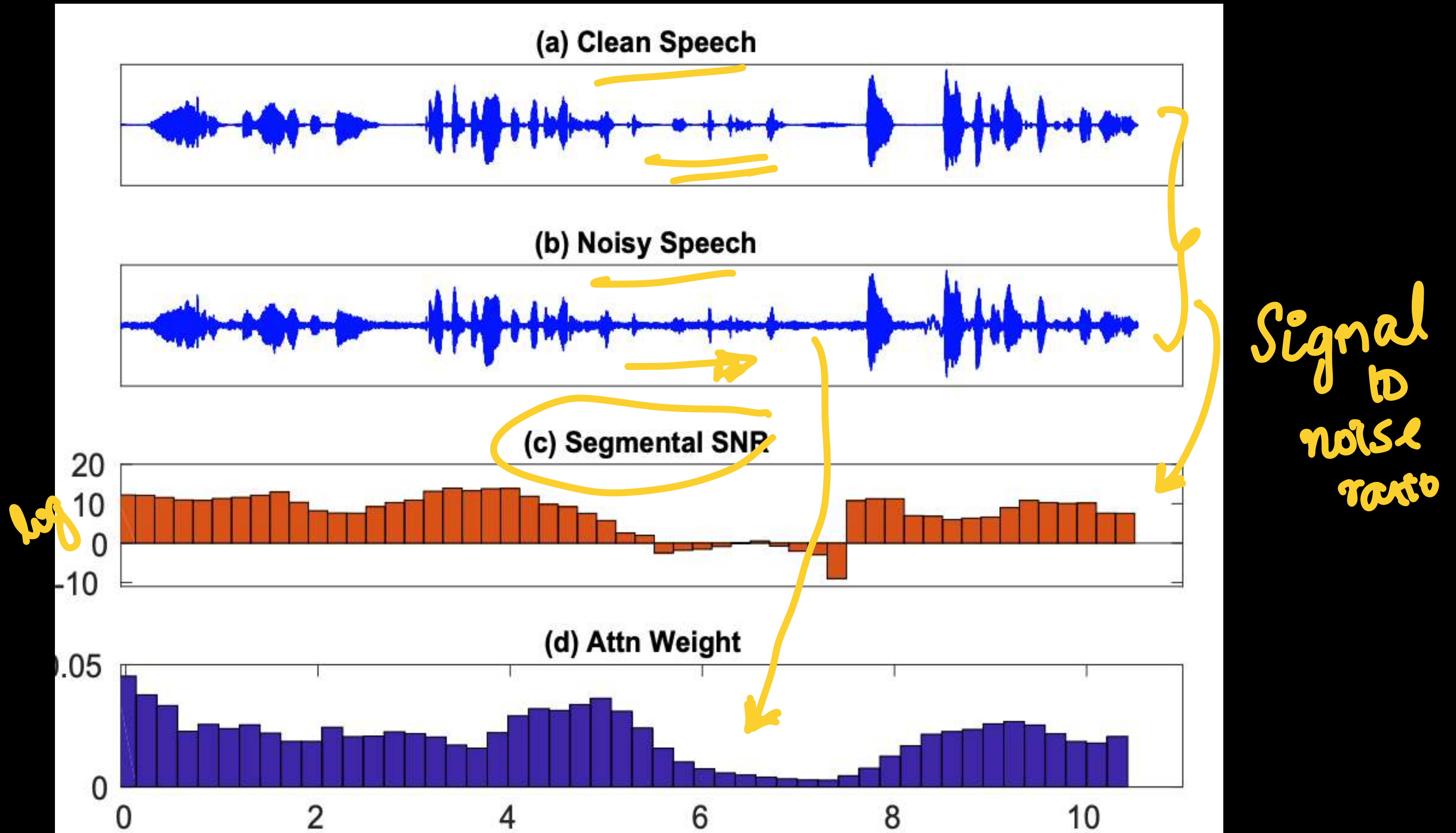
4s - 9s I couldn't scream, I couldn't shout, I couldn't
even move my arms up, or my legs

9s -11s

Using attention mechanism for explainability



Using attention mechanism for visualization



Summary thus far

✳ Analyzing trained neural networks

- ✓ Hierarchical representations
- ✓ Activation maps to determine saliency
- ✓ Incorporating attention mechanism for improved explainability

